

Co I SHORT ANSWER

1. CCD vs CMOS Image Acquisition Systems

- CCD: Better image quality, low noise, high sensitivity, but expensive and consumes more power.
- CMOS: Faster, lower power consumption, cheaper, but may have slightly more noise.

Support: CCD is preferred for scientific imaging, while CMOS is widely used in cameras + mobile devices.

2. Causes of Image Distortion + Improvements.

Causes:

- Poor lighting
- Lens defects
- Incorrect camera alignment
- Sensor noise.

Improvements:

- Use proper illumination
- Calibrate the camera
- Use quality lenses
- Apply image correction techniques.

3. Computer Vision Features in Automated Surveillance.

- Object detection identifies people + vehicles
- Motion detection tracks movement
- Face recognition identifies individuals
- Real-time monitoring ensures security + safety.

4. Digital Image Fundamentals in Satellite Image Analysis

- High resolution images provide more detail.
- Accurate image acquisition reduces error.
- Proper sampling and quantization improve analysis accuracy.
- Better image quality leads to reliable land classification.

5. Effect of Sampling, Quantization & Acquisition on OCR

- Good image acquisition captures clear text.
- Proper sampling preserves character details.
- Adequate quantization maintains contrast.
- Together they improve character recognition.

6. Optimizing Storage and Processing Time

- Reduce sampling rate slightly while keeping important details.
- Use moderate quantization levels.
- Apply image compression techniques.
- This reduces storage w/o major quality loss.

Uneven Lighting & Poor Contrast

- Uneven illumination causes brightness variations
- Shadows & highlights reduce visibility
- Use uniform lighting
- Apply contrast enhancement

8. Quantization in Medical Imaging

- Low quantization levels may remove subtle intensity differences
- Important diagnostic details can be lost
- Higher intensity resolution preserves critical information

9. Image Acquisition and CV in Self-driving Cars

- Cameras capture road & traffic info
- Computer vision detects lane, signs & obstacles
- Real-time image processing supports safe navigation
- Accurate acquisition improves driver's safety

10. Choosing Sampling Rate & Intensity Resolution

- Use a high sampling rate to capture fine details
- Use high intensity resolution to represent subtle brightness changes
- This improves image clarity & feature detection